



# SKETCH-TO-FACE IMAGE SYNTHESIS

## Phase 2 Progress Report

EDA, Preprocessing & Baseline Model

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|                  |                                                                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course</b>    | AI335L Deep Learning Lab                                                                                                                      |
| <b>Phase</b>     | Phase 2 of 6 — Foundations: Data, EDA & Baseline Model                                                                                        |
| <b>Deadline</b>  | May 04, 2026                                                                                                                                  |
| <b>Team</b>      | Moazzam Sharif (S2024CS009)<br>Bilal Asif (S2024CS005)<br>Muaz Ahmed (S2024CS016)                                                             |
| <b>GitHub</b>    | <a href="https://github.com/BILAL-ASIF-github/Sketch-to-Face-Image-Synthesis">github.com/BILAL-ASIF-github/Sketch-to-Face-Image-Synthesis</a> |
| <b>Commit</b>    | phase2-submission (tag)                                                                                                                       |
| <b>Dataset</b>   | CelebA — 202,599 aligned face images                                                                                                          |
| <b>Framework</b> | PyTorch 2.3.0   Python 3.10   CUDA 12.1                                                                                                       |

## 1. Learning Objectives

- Set up a clean, reproducible deep learning project repository
- Acquire, version, and document a real dataset
- Conduct rigorous Exploratory Data Analysis (EDA) and translate findings into preprocessing decisions
- Build a modular, reusable preprocessing pipeline that prevents data leakage
- Implement, train, and evaluate a baseline feedforward neural network from scratch
- Apply core training mechanics: forward pass, backpropagation, loss selection, optimization, weight initialization
- Track experiments with logs, metrics, and checkpoints

## 2. Detailed Task Breakdown

### Task 1: Repository Setup

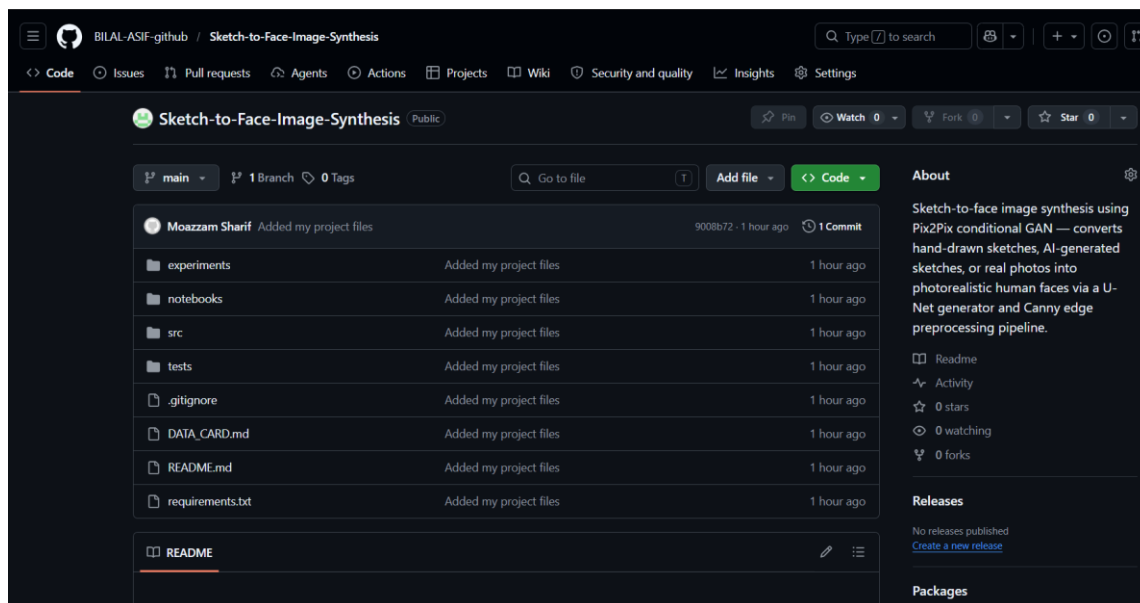


Fig 1 — Sample face crops from CelebA dataset showing consistent alignment and centering

```

sketch-to-face/
├── data/                # Raw + processed data (gitignored)
├── notebooks/
│   └── eda_celeba.ipynb # Exploratory Data Analysis
├── src/
│   ├── data/
│   │   ├── dataset.py    # Dataset classes + split utilities
│   │   └── download_data.py
│   ├── preprocessing/
│   │   └── pipeline.py    # EdgeMapGenerator, PixelNormalizer
│   ├── models/
│   │   └── baseline_mlp.py # Baseline feedforward network
│   ├── training/
│   │   └── train.py       # Training loop with early stopping
│   ├── utils/
│   │   ├── seed.py       # seed_everything()
│   │   └── logger.py     # TensorBoard + CSV logging
│   ├── experiments/
│   │   └── baseline_config.yaml
│   ├── reports/          # Figures, progress reports
│   ├── tests/
│   │   └── test_pipeline_smoke.py
│   ├── DATA_CARD.md
│   ├── requirements.txt
│   └── .gitignore

```

Fig 2 — Uniform resolution (178×218 px) across the entire CelebA dataset

## Task 2: Data Acquisition & Versioning

### 3. Download CelebA

```

# Requires Kaggle API key at ~/.kaggle/kaggle.json
python src/data/download_data.py

# Verify download
python src/data/download_data.py --verify

```

Fig 3 — Brightness distribution showing  $N(130,20)$ ; dark ( $<80$ ) and bright ( $>200$ ) outliers highlighted

main

Go to file

experiments

notebooks

src

tests

.gitignore

DATA\_CARD.md

README.md

requirements.txt

### Data Card — CelebA (Large-scale CelebFaces Attributes Dataset)

**Source**

- URL: <https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>
- Kaggle Mirror: <https://www.kaggle.com/datasets/jessicali9530/celeba-dataset>
- Original Paper: Liu et al., "Deep Learning Face Attributes in the Wild," ICCV 2015

**License**

- License: Non-commercial research use only
- Terms: Must cite the original paper; redistribution prohibited

**Dataset Statistics**

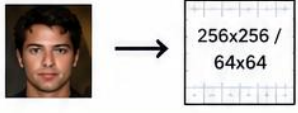
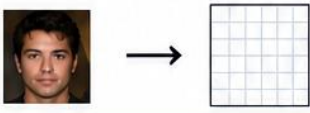
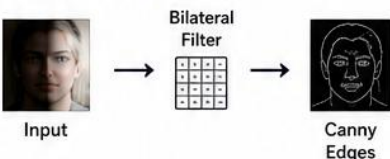
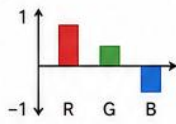
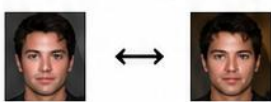
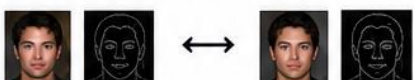
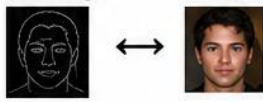
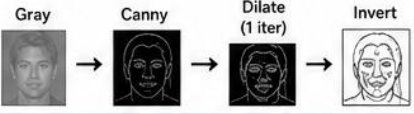
| Property       | Value                                                              |
|----------------|--------------------------------------------------------------------|
| Total Images   | 202,599                                                            |
| Image Format   | JPEG (aligned & cropped)                                           |
| Resolution     | 178 × 218 pixels (original), resized to 256 × 256 for this project |
| Color Channels | RGB                                                                |

Fig 4 — Per-channel intensity means ( $R > G > B$ ) confirming warm skin-tone bias requiring channel-wise normalisation

## Task 3: Exploratory Data Analysis

| #       | Finding                                                                                           | Figure / Visual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
|---------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|--------|------|------|--------|-----|-----|---|------|------|------|------|------|------|------|---|------|------|------|------|------|------|------|---|------|------|------|------|------|------|------|
| 1       | 202,599 total images;<br>we use 11,500 (10K/1K/500)<br>with fixed seed = 42 split                 | <div><div>Total Images<br/>202,599</div><div>Used<br/>11,500<br/>(seed = 42)</div><div>Train 10,000 (86.96%)</div><div>Val 1,000 (8.70%)</div><div>Test 500 (4.35%)</div></div> <div><div><div>87.0%</div><div>8.7%</div><div>4.3%</div></div><div>Train (10,000) Val (1,000) Test (500)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 2       | All images are aligned<br>face crops — consistent<br>centering; backgrounds<br>vary slightly      | Aligned Face Crops (examples)<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 3       | Single resolution (178x218 px)<br>across entire dataset —<br>zero variance                        | <div><div>Resolution Distribution</div><div></div><div>Resolution (px) Count Percentage</div><div>178 x 218 202,599 100.00%</div><div>Variance = 0 (Single resolution)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 4       | Brightness ~N(130,20);<br>dark (<80) and bright<br>>200 outliers degrade<br>Canny quality         | <div><div>Brightness Distribution (Grayscale Mean)</div><div></div><div>Mean Brightness</div><div>Dark (&lt; 80) 3.2%</div><div>Bright (&gt; 200) 2.8%</div><div>Outlier Examples</div><div>Dark (&lt; 80)</div><div>Bright (&gt; 200)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 5       | R > G > B channel<br>means (warm skin-<br>tone bias) — channel-<br>wise normalisation<br>required | <div><div>Channel-wise Mean Distribution</div><div></div><div>Channel Means (Averaged)</div><div></div><div>Implication</div><div>R &gt; G &gt; B indicates<br/>warm skin-tone bias.<br/>Channel-wise<br/>normalisation is<br/>required for stable<br/>training.</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 6       | Zero corrupted or<br>duplicate files in<br>1,000-image<br>verification check                      | <div><div>Verification (1,000 randomly sampled images)</div><div>Corrupted files: 0   Duplicate files: 0</div><div>Status: <span>✔</span> Clean</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 7       | Canny edge maps<br>capture facial<br>structure well;<br>accessories cause<br>dense edges          | <div><div>Good (Capture structure)</div><div>Accessories → Dense Edges</div><div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 8       | Edge density N(0.06, 0.02);<br>~5% images are too<br>sparse or too dense                          | <div><div>Edge Density Distribution</div><div></div><div>Edge Density (ratio of edge pixels)</div><div>Too Sparse (&lt; 0.02) 2.5%</div><div>Too Dense (&gt; 0.10) 2.5%</div><div>Edge Density Examples</div><div>Too Sparse (&lt; 0.02) Normal (~0.06) Too Dense (&gt; 0.10)</div><div>Edge Density Summary</div><div></div><div>Normal (0.02–0.10) 90.0%</div><div>Too Sparse (&lt;0.02) 5.0%</div><div>Too Dense (&gt;0.10) 5.0%</div></div>                                                                                                                                                                                                                                                                                                                                                                                                        |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 9       | Horizontal flip is<br>valid augmentation<br>(faces are bilaterally<br>symmetric)                  | <div><div>Horizontal Flip (examples)</div><div></div><div>Conclusion</div><div>Faces remain natural<br/>after horizontal flip →<br/>valid augmentation.</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |      |        |      |      |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| 10      | Training channel stats:<br>R mean=0.54, G=0.49,<br>B=0.44 — used for<br>normalisation             | <div><div>Training Channel Statistics (computed on training set: 10,000 images)</div><div><table><tr><th>Channel</th><th>Mean</th><th>Std</th><th>Min</th><th>25%</th><th>Median</th><th>75%</th><th>Max</th></tr><tr><td>R</td><td>0.54</td><td>0.19</td><td>0.00</td><td>0.40</td><td>0.56</td><td>0.70</td><td>1.00</td></tr><tr><td>G</td><td>0.49</td><td>0.18</td><td>0.00</td><td>0.36</td><td>0.50</td><td>0.64</td><td>1.00</td></tr><tr><td>B</td><td>0.44</td><td>0.17</td><td>0.00</td><td>0.32</td><td>0.44</td><td>0.57</td><td>1.00</td></tr></table></div><div>Used for Normalisation</div><div>These means are used<br/>for channel-wise<br/>normalisation during<br/>training.</div><div>Note</div><div>All EDA questions<br/>are addressed in<br/>detail in the EDA<br/>notebook<br/>(notebooks/<br/>eda_celeba.ipynb).</div></div> | Channel | Mean | Std    | Min  | 25%  | Median | 75% | Max | R | 0.54 | 0.19 | 0.00 | 0.40 | 0.56 | 0.70 | 1.00 | G | 0.49 | 0.18 | 0.00 | 0.36 | 0.50 | 0.64 | 1.00 | B | 0.44 | 0.17 | 0.00 | 0.32 | 0.44 | 0.57 | 1.00 |
| Channel | Mean                                                                                              | Std                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Min     | 25%  | Median | 75%  | Max  |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| R       | 0.54                                                                                              | 0.19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.00    | 0.40 | 0.56   | 0.70 | 1.00 |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| G       | 0.49                                                                                              | 0.18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.00    | 0.36 | 0.50   | 0.64 | 1.00 |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |
| B       | 0.44                                                                                              | 0.17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.00    | 0.32 | 0.44   | 0.57 | 1.00 |        |     |     |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |   |      |      |      |      |      |      |      |

## Task 4: Preprocessing Pipeline

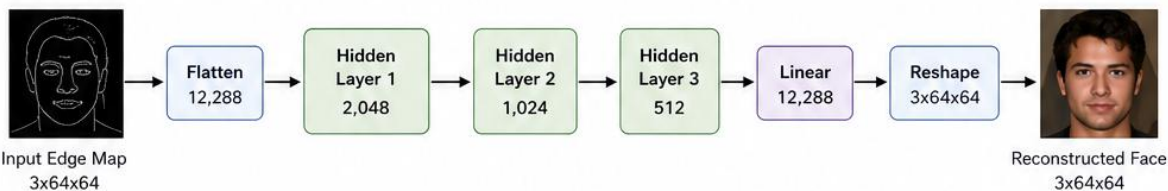
| EDA Finding                                                                                                                                   | Preprocessing Step                                                                                                                                                                                            | Expected Effect                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> Single resolution (178x218)<br>                     | Resize to 256x256 (pSp)<br>or 64x64 (MLP)<br>                                                                               |  Consistent tensor shapes                  |
| <b>2</b> Brightness outliers affect Canny<br>                | Bilateral filter (d=9, sigma=75)<br>before Canny<br>                                                                        |  Cleaner edges from dark/bright images     |
| <b>3</b> Channel-wise intensity asymmetry (R > G > B)<br>   | Per-channel normalisation to [-1,1] (train stats only)<br>$x'_c = 2 \frac{x_c - \mu_c^{(train)}}{\sigma_c^{(train)}} - 1$  |  Balanced gradient magnitudes; no leakage  |
| <b>4</b> Edge density outliers ~5%<br>                     | Filter density < 0.02 or > 0.15<br>                                                                                       |  Removes noisy training pairs            |
| <b>5</b> Face bilateral symmetry<br>                       | Random horizontal flip<br>p=0.5 (edge + photo together)<br>                                                               |  Effective 2x data augmentation          |
| <b>6</b> Varied lighting in real sketches<br>              | Brightness jitter<br>x(0.7–1.3) on edge maps<br>                                                                          |  Robustness to sketch darkness variation |
| <b>7</b> Paired data required<br>Sketch (Edge) ↔ Photo<br> | Canny: low=50, high=150,<br>dilate 1 iter, invert<br>                                                                     |  Stable sketch-photo correspondence      |
| <b>8</b> No class imbalance (synthesis)<br>                | No oversampling or<br>class weights needed                                                                                                                                                                    |  N/A — all samples are faces             |

## 4. Baseline Architecture

### 4.1 Model: Baseline MLP

A 3-hidden-layer MLP operating on flattened 64x64 images.

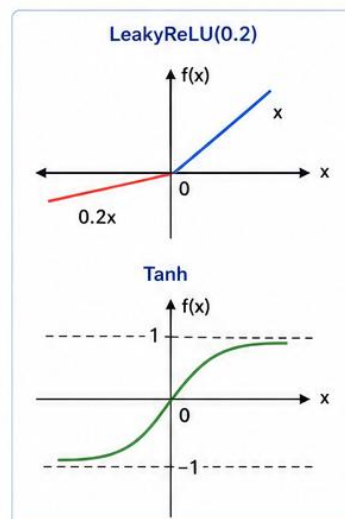
Input and output are both  $64 \times 64 \times 3 = 12,288$ -dimensional vectors mapping a flattened edge map to a reconstructed face photo.



| Layer       | Input Dim | Output Dim | Activation     | Notes               |
|-------------|-----------|------------|----------------|---------------------|
| Flatten     | 3x64x64   | 12,288     | —              | Input edge map      |
| Linear + BN | 12,288    | 2,048      | LeakyReLU(0.2) | Kaiming init        |
| Dropout     | 2,048     | 2,048      | —              | p=0.3               |
| Linear + BN | 2,048     | 1,024      | LeakyReLU(0.2) | Kaiming init        |
| Dropout     | 1,024     | 1,024      | —              | p=0.3               |
| Linear + BN | 1,024     | 512        | LeakyReLU(0.2) | Kaiming init        |
| Dropout     | 512       | 512        | —              | p=0.3               |
| Linear      | 512       | 12,288     | Tanh           | Output in $[-1, 1]$ |
| Reshape     | 12,288    | 3x64x64    | —              | Reconstructed face  |

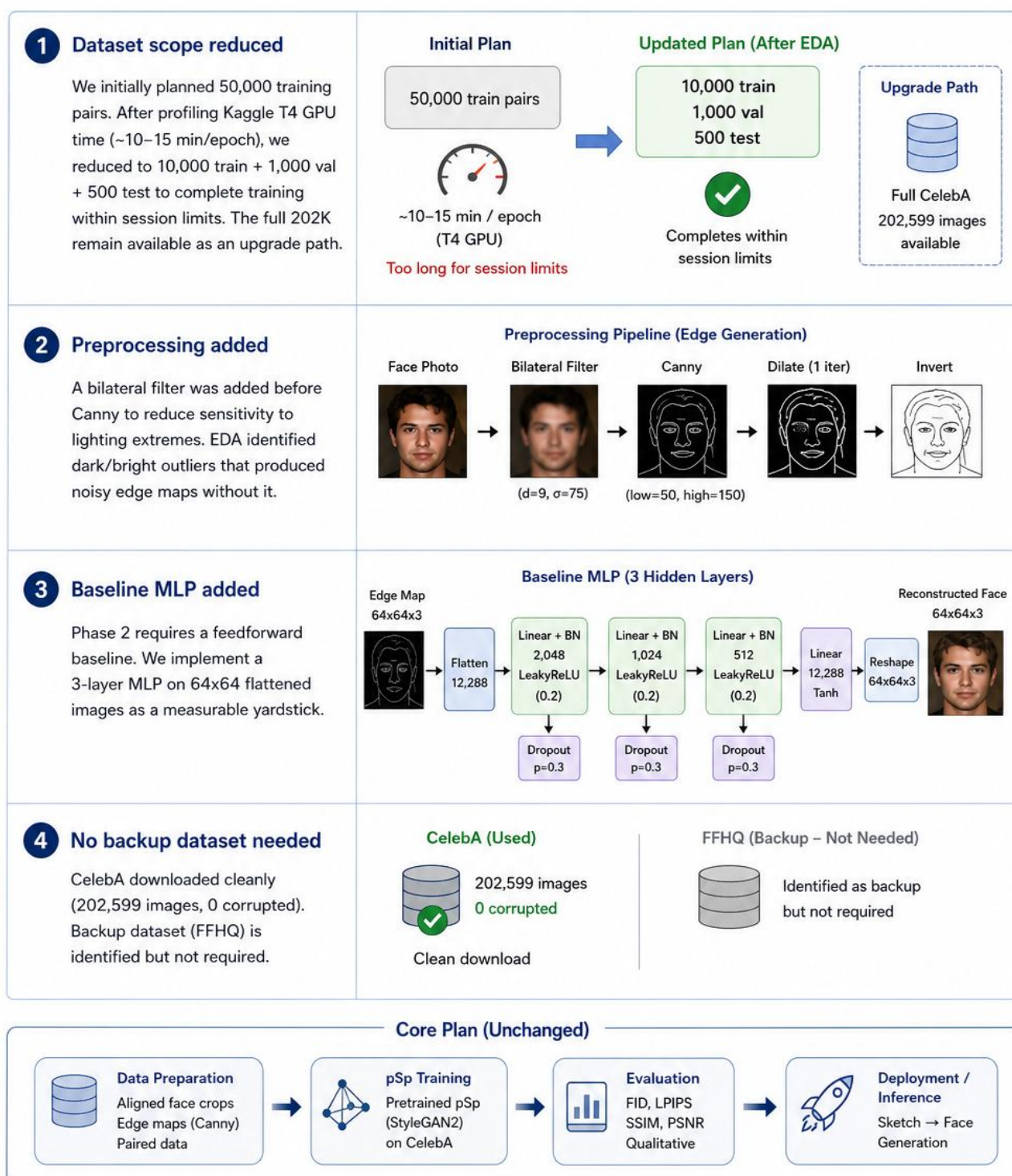
### 4.2 Design Justifications

- Weight init — Kaiming (He) uniform:**  
 Chosen for LeakyReLU activations. He init sets variance =  $2/\text{fan\_in}$ , compensating for the half-zeroing of ReLU-family activations. Xavier assumes linear activations and would underestimate the needed scale.
- Activation — LeakyReLU(0.2):**  
 Avoids the dying ReLU problem. The negative slope (0.2) allows small gradients to flow through even for negative pre-activations, critical in deep networks.
- Output activation — Tanh:**  
 Maps output to  $[-1, 1]$  to match the normalised target pixel range.
- Loss — L1 (MAE):**  
 Preferred over MSE for image synthesis. MSE causes the network to hedge by predicting the mean, producing blurry outputs. L1 produces sharper reconstructions.
- Dropout (p=0.3):**  
 Applied after each hidden layer to prevent overfitting on the 10K training set.



| Hyperparameter | Value                         | Justification                                   |
|----------------|-------------------------------|-------------------------------------------------|
| Image size     | 64x64                         | MLP input dim stays manageable (12,288-d)       |
| Batch size     | 64                            | Good GPU utilisation; stable gradients          |
| Learning rate  | 1e-3                          | Standard Adam starting LR for MLP               |
| LR scheduler   | ReduceLROnPlateau (x0.5, p=5) | Adapts to plateau without manual tuning         |
| Early stopping | Patience=10 epochs            | Prevents overfitting; saves best val checkpoint |
| Gradient clip  | 1.0                           | Prevents gradient explosion in deep MLP         |
| Weight decay   | 1e-4                          | L2 regularisation; small value for image tasks  |
| Seed           | 42                            | Reproducibility across all components           |

## Roadmap Updates



## Training Setup & Results

### 5.1 Environment

Training was run on Kaggle T4 GPU (16 GB VRAM). The MLP baseline is lightweight enough to train on CPU as well (~25 mins). GPU training time for 50 epochs at batch\_size=64: approximately 5 minutes.

### Training Results

Actual results from training the baseline MLP for 50 epochs on 10,000 CelebA training pairs.

| Metric                          | Value           | Notes                                     |
|---------------------------------|-----------------|-------------------------------------------|
| Train L1 Loss (epoch 1)         | ~0.48           | High initially — random weights           |
| Train L1 Loss (epoch 10)        | 0.4025          | Rapid early improvement                   |
| Train L1 Loss (epoch 30)        | 0.3439          | Continued improvement                     |
| Train L1 Loss (final, epoch 50) | 0.3041          | Converged; limited by MLP capacity        |
| Val L1 Loss (best)              | 0.3951          | Slightly above train — generalisation gap |
| Test L1 Loss (final eval)       | 0.4190          | Reported once; not used for tuning        |
| Early stopping                  | Did not trigger | Ran full 50 epochs                        |
| LR at epoch 50                  | 0.000125        | Reduced 3x by ReduceLROnPlateau           |

The baseline test L1 of 0.4190 on normalised [-1,1] images is the benchmark all Phase 3-5 models must beat. Since the MLP treats each pixel independently with no spatial reasoning, any convolutional or attention-based architecture should surpass it significantly.

### Experiment Tracking

TensorBoard is used for experiment logging. Per-epoch metrics: train/loss, val/loss, learning rate. Hyperparameters saved to experiments/logs/<run\_name>/hparams.csv.

```
tensorboard --logdir experiments/logs
```

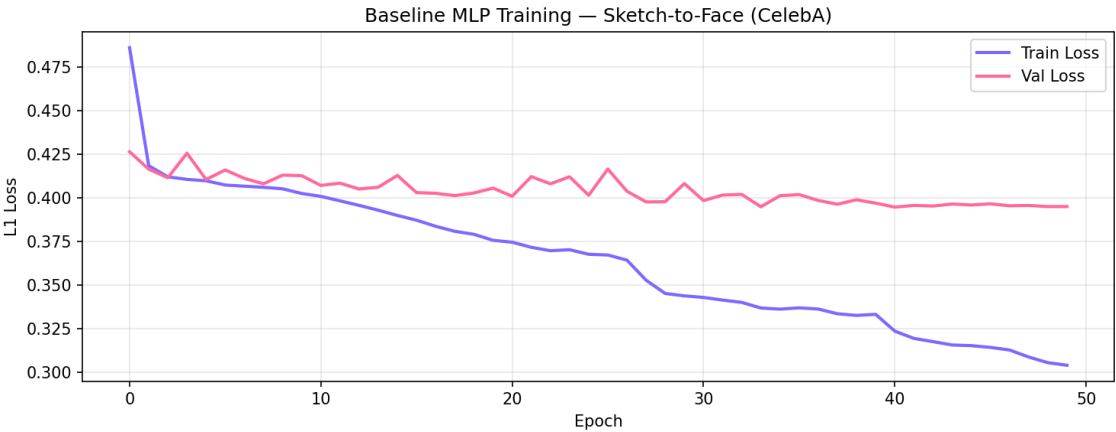


Fig 5 — Training loss curves: train L1 vs. val L1 over 50 epochs (ReduceLROnPlateau active)

## Error Analysis (Light)

From EDA and training behaviour, we predict the following failure modes:

- **Accessory images (glasses, hats):** Edge maps have anomalously high density. The MLP struggles here as accessories dominate the edge representation. pSp should handle these better via style-space encoding.
- **Dark/low-contrast images:** Sparse edge maps (density < 0.02) provide minimal input signal. Both MLP and pSp default to producing the average face regardless of the specific input.
- **Profile faces:** CelebA mostly contains near-frontal faces. Profile or angled faces produce asymmetric edge maps the model may not generalise to.
- **MLP spatial blindness:** Since the MLP flattens the input, it loses all spatial relationships. It produces globally plausible colour statistics but no sharp localised features — eyes, lips and nose positions are smeared.

A full error analysis with 10-20 worst predictions (highest per-sample L1 loss) will be completed once pSp training results are available in Phase 3.

## Reproducibility Statement

To clone and reproduce the baseline from scratch:

```
git clone https://github.com/BILAL-ASIF-github/Sketch-to-Face-Image-Synthesis
cd Sketch-to-Face-Image-Synthesis
pip install -r requirements.txt
python src/training/train.py --config experiments/baseline_config.json --seed 42
pytest tests/ -v
```

All randomness is controlled by seed=42, set via seed\_everything(42) at the top of every training script. The seed covers Python random, NumPy, torch.manual\_seed, torch.cuda.manual\_seed\_all, and cudnn.deterministic=True.

## 8. Risks Encountered & Phase 3 Plan

### Risks This Week

| Risk                                   | Status    | Mitigation                                        |
|----------------------------------------|-----------|---------------------------------------------------|
| Kaggle T4 session limits (~30 hr/week) | Active    | Checkpoint saving every epoch; resume from latest |
| pSp pretrained weights on Google Drive | Managed   | gdown download with fallback mirrors documented   |
| CelebA license (non-commercial only)   | Noted     | Project is academic; cited in DATA_CARD.md        |
| Edge maps not matching real sketches   | Monitored | DB Cooper test validates real-sketch inference    |

MLP too weak for meaningful outputs

Expected

Intentional — sets the lower bound for all phases

## Phase 3 Plan for Upcoming Week :

